Changed occupancy feature

Please note that our system returns rates with a changed occupancy. These options appear whenever the system generates a rate that does not exactly match the searched criteria. You must take this into account and ensure that your application correctly interprets and processes such rates, so that the corresponding occupancy elements are accurately passed in the confirmBooking request. This will allow successful booking of rates with a changed occupancy.
Rates with changed occupancy are one of the key characteristics introduced starting from **API version V3**. If you choose to filter them out under the assumption that they do not meet the user's search criteria, please note that this is not correct — these rates fully accommodate the requested occupancy. Otherwise, the system would not return them as available.
For example, a changed occupancy rate might:

- Include an **extra bed**, allowing a child or adult to be accommodated.
- Apply a rule where a **child exceeding the hotel's defined age range** is treated as an adult.

Our system provides multiple valid rate options for the user's requested criteria.
In the following section, we will provide examples explaining how to correctly interpret and populate all changed occupancy–related elements.
Below, you can find several examples illustrating various **changed occupancy rate scenarios**.

Let assume we are searching for 1 room with the below occupancy:

```
<room runno="0">
    <adultsCode>2</adultsCode>
    <children no="3">
        <child runno="0">2</child>
        <child runno="1">6</child>
        <child runno="2">12</child>
    </children>
    <rateBasis>-1</rateBasis>
    <passengerNationality>66</passengerNationality>
    <passengerCountryOfResidence>88</passengerCountryOfResidence>
</room>
```

The system might return as an option the below rate with a changedOccupancy:

```
<roomType runno="0" roomtypecode="165902">
    <name>TWO BEDROOM SUITE</name>
    <twin>no</twin>
    <roomInfo>
        <maxOccupancy>6</maxOccupancy>
        <maxAdultWithChildren>4</maxAdultWithChildren>
        <minChildAge>0</minChildAge>
        <maxChildAge>11</maxChildAge>
        <maxAdult>4</maxAdult>
        <maxExtraBed>0</maxExtraBed>
        <maxChildren>5</maxChildren>
    </roomInfo>
    <specials count="1">
        <special runno="0">
            <type>stayXGetDiscountPromotion</type>
            <specialName>25 % Off</specialName>
            <discount>25</discount>
            <stay>1</stay>
        </special>
    </specials>
    <rateBases count="3">
        <rateBasis runno="0" id="0" description="Room Only">
            <validForOccupancy>
                <adults>3</adults><!-- because the system identified the fact that this roomType has a child age
                restriction of 1 to 11 it has transformed the 12 years old child in an additional adult-->
                <children>2</children>
                <childrenAges>2, 6</childrenAges>
                <extraBed>0</extraBed>
                <extraBedOccupant>adult</extraBedOccupant>
            </validForOccupancy>
            <changedOccupancy>3,2,2_6,0</changedOccupancy>
```

Therefore, in the **confirmBooking** request the occupancy will have to be passed as below:

```
<room runno="1">
    <roomTypeCode>165902</roomTypeCode>
    <selectedRateBasis>0</selectedRateBasis>
    <allocationDetails>...............</allocationDetails>
    <adultsCode>3</adultsCode><!-- in case of a rate with changed occupancy this element will be
    populated with the value returned in the validForOccupancy node in the adults element.-->
    <actualAdults>2</actualAdults><!-- this element will always be populated with the initial
    searched values-->
    <children no="2"><!-- in case of a rate with changed occupancy this element will be populated
    with the value returned in the validForOccupancy node in the children and childrenAges
    elements.-->
        <child runno="0">2</child>
        <child runno="1">6</child>
    </children>
    <actualChildren no="3"><!-- this node will always be populated with the initial searched
    values-->
        <actualChild runno="0">2</actualChild>
        <actualChild runno="1">6</actualChild>
        <actualChild runno="2">12</actualChild>
    </actualChildren>
    <extraBed>0</extraBed> <!-- in case of a rate with changed occupancy this element will be
    populated with the value returned in the validForOccupancy node in the extrabed element.-->
    <passengerNationality>66</passengerNationality>
    <passengerCountryOfResidence>88</passengerCountryOfResidence>
    <passengersDetails>
        <passenger leading="yes">
            <salutation>147</salutation>
            <firstName>John</firstName>
            <lastName>Doe</lastName>
        </passenger>
    </passengersDetails>
    </room>
```

Let's take another example:
I have performed a search with the below occupancy:

```
<rooms no="2">
    <room runno="0">
        <adultsCode>3</adultsCode>
        <children no="0"/>
        <rateBasis>-1</rateBasis>
        <passengerNationality>81</passengerNationality>
        <passengerCountryOfResidence>100</passengerCountryOfResidence>
    </room>
    <room runno="1">
        <adultsCode>2</adultsCode>
        <children no="2">
            <child runno="0">2</child>
            <child runno="1">5</child>
        </children>
        <rateBasis>-1</rateBasis>
        <passengerNationality>81</passengerNationality>
        <passengerCountryOfResidence>100</passengerCountryOfResidence>
    </room>
</rooms>
```

From the returned results I have selected and successfully blocked the following rates with changedOccupancy:

- for room runno 0:

```
<rateBasis runno="1" id="1331" description="Breakfast">
    <validForOccupancy>
        <adults>3</adults>
        <extraBed>1</extraBed>
        <extraBedOccupant>adult</extraBedOccupant>
    </validForOccupancy>
    <changedOccupancy>3,0,,1</changedOccupancy>
    <status>checked</status>
```

- for room runno 1:

```
<rateBasis runno="1" id="1331" description="Breakfast">
    <validForOccupancy>
        <adults>2</adults>
        <children>1</children>
        <childrenAges>2</childrenAges>
        <extraBed>1</extraBed>
        <extraBedOccupant>child</extraBedOccupant>
    </validForOccupancy>
    <changedOccupancy>2,1,2,1</changedOccupancy>
    <status>checked</status>
```

For a successful booking the changed occupancy returned in the getRooms should have been passed in the confirm-Booking as follows:

```
<room runno="0">
    <roomTypeCode>59574</roomTypeCode>
    <selectedRateBasis>1331</selectedRateBasis>
    <allocationDetails/>
    <adultsCode>3</adultsCode><!-- this element must be populated with the value of the <adults> element from the
    validForOccupancy node-->
    <actualAdults>3</actualAdults><!-- initial searched value-->
    <children no="0"/><!--this node be populated with the values of the <children> and <childrenAges> elements from the
    validForOccupancy node-->
    <actualChildren no="0"/><!--  initial searched value-->
    <extraBed>1</extraBed><!--this element must be populated with the value of the <extraBed> element from the validForOccupancy
    node-->
    <passengerNationality>81</passengerNationality>
    <passengerCountryOfResidence>100</passengerCountryOfResidence>
    <passengersDetails>
</room>
```

.....

```
<room runno="1">
    <roomTypeCode>59574</roomTypeCode>
    <selectedRateBasis>1331</selectedRateBasis>
    <allocationDetails>...........</allocationDetails>
    <adultsCode>2</adultsCode><!-- this element must be populated with the value of the <adults> element from the
    validForOccupancy node-->
    <actualAdults>2</actualAdults><!-- initial searched value-->
    <children no="1"><!--  this node must be populated with the values of the <children> and <childrenAges> elements from the
    validForOccupancy node-->
        <child runno="0">2</child>
    </children>
    <actualChildren no="2> <!-- initial searched value-->
        <actualChild runno="0">2</actualChild>
        <actualChild runno="1">5</actualChild>
    </actualChildren>
    <extraBed>1</extraBed><!--this element must be populated with the value of the <extraBed> element from the validForOccupancy
    node-->
    <passengerNationality>81</passengerNationality>
    <passengerCountryOfResidence>100</passengerCountryOfResidence>
    <passengersDetails>
</room>
```

In case of rates that don't have a changed occupancy the adultsCode/actualAdults and children/actualChildren will be passed in the confirmBooking with the same value requested in the initial search.

If you are not willing to implement the changed occupancy Feature, it is mandatory for you to filter these rates and not show them to your customers. You can implement a validation based on validForOccupancy/changedOccupancy elements.